

Capital Budgeting and Infrastructure Project Valuation with Monte Carlo Risk Simulation

Project Domain: Clean Infrastructure & Energy Finance	Target Asset: 100MW Solar Park / 50MW Wind Farm
Data Source: Kaggle / The World Bank PPI Database	Platform: Python, Streamlit, Plotly, SciPy, PyTorch
Primary Methodology: Multi-Stage DCF & WACC Hurdle Rate	Risk Simulation: 5,000-Iteration Stochastic Monte Carlo

1. Executive Overview & Real-World Motivation

Large-scale clean infrastructure assets—such as 100MW Solar PV Power Plants, 50MW Onshore Wind Farms, and Clean Water Desalination Facilities—require **multi-million-dollar upfront capital expenditures (\$20M to \$100M+)** and operate across 20-to-30-year concession lifecycles. Institutional lenders, private equity funds, and Chief Financial Officers (CFOs) face substantial financial exposure due to long payback periods and exposure to power price fluctuations.

The Core Financial Engineering Dilemma: A single static Net Present Value (NPV) calculation assumes fixed electricity selling tariffs and zero construction delays over 20 years. In reality, energy market price volatility, supply chain cost overruns, and macroeconomic inflation create significant downside tail-risk. This project provides an interactive decision support system combining **Deterministic Capital Budgeting** (NPV, IRR, PI, Payback, and Levelized Cost of Energy - LCOE) with a **5,000-iteration Stochastic Monte Carlo Risk Engine**.

2. Public Dataset Architecture & Attribute Dictionary

The project integrates verified public data from **The World Bank Private Participation in Infrastructure (PPI) Database** and **IRENA Global Renewable Cost Benchmarks** (hosted on Kaggle). The key parameters used by the engine include:

Column Name	Financial Definition	Baseline Value	Role in Valuation
Project_Name	Asset name and technology type	100MW Solar Park	Identifier
CapEx_USD_M	Initial construction and land cost	\$30.0 Million	Year 0 cash outflow (\$I_0\$)
Annual_Gen_MWh	Total annual electricity generated	180,000 MWh	Volume driver for revenue
Base_Tariff_USD	Contracted PPA tariff per MWh	\$48.00 / MWh	Price driver for revenue
Lifetime_Years	Asset operational lifecycle	20 Years	Discounting horizon (\$N\$)
Annual_OpEx_M	Maintenance, labor, grid fees	\$1.50 Million/yr	Operating cash outflow
WACC_Percent	Weighted Average Cost of Capital	8.00%	Hurdle rate / discount rate
Degradation_Pct	Annual efficiency loss of panels	0.50% / year	Annual generation decay

3. Core Financial Engineering Mathematical Formulations

A. Annual Net Cash Flow & Discounted Cash Flow (DCF)

For each operational year t (from 1 to N), annual generation decays by the degradation factor, and operating expenses escalate by annual inflation i :

$$\begin{aligned} \text{Revenue}_t &= [\text{Annual Generation} \times (1 - \text{Degradation})^{t-1} \times \text{Tariff}_t] / 1,000,000 \\ \text{OpEx}_t &= \text{Base OpEx} \times (1 + \text{Inflation})^{t-1} \\ \text{Net Cash Flow (CF}_t) &= (\text{Revenue}_t - \text{OpEx}_t) \times (1 - \text{Corporate Tax Rate}) \end{aligned}$$

B. Net Present Value (NPV)

NPV quantifies the aggregate net economic value created in today's dollars by discounting future cash flows at WACC:

$$\text{NPV} = -I_0 + \sum_{t=1}^N [\text{CF}_t / (1 + \text{WACC})^t]$$

Decision Rule: Accept if $\text{NPV} > 0$ (Value Accretive); Reject if $\text{NPV} \leq 0$ (Value Destructive).

C. Internal Rate of Return (IRR) & Levelized Cost of Energy (LCOE)

$$\begin{aligned} 0 &= -I_0 + \sum_{t=1}^N [\text{CF}_t / (1 + \text{IRR})^t] \quad (\text{Solved via Brent's Numerical Root Algorithm}) \\ \text{LCOE (\$/MWh)} &= [\text{Initial CapEx} + \sum (\text{OpEx}_t / (1 + \text{WACC})^t)] / [\sum (\text{Generation}_t / (1 + \text{WACC})^t)] \end{aligned}$$

LCOE represents the exact minimum price floor per unit of electricity needed to recover all capital and operating costs.

4. Stochastic Monte Carlo Risk Engine & Sensitivity Analysis

To evaluate real-world uncertainty, the platform incorporates a **5,000-iteration Monte Carlo simulation**. In each simulation run, three independent stochastic variables are simultaneously sampled:

Stochastic Variable	Probability Distribution	Mathematical Specification
PPA Electricity Tariff	Gaussian / Normal	$Tariff \sim N(\mu = \$48.0, \sigma = 15\%)$
CapEx Construction Cost	Uniform Overrun	$CapEx \sim U(1.00 \times I_0, 1.25 \times I_0)$
Macroeconomic Inflation	Gaussian / Normal	$Inflation \sim N(\mu = 2.5\%, \sigma = 1.0\%)$

Risk Quantification Metrics Generated:

- **Probability of Loss:** $P(NPV < 0) = (\text{Count of negative NPV iterations} / 5,000) \times 100\%$
- **95% Value-at-Risk (VaR):** The 5th percentile worst-case simulated NPV threshold.
- **Conditional VaR (CVaR / Expected Shortfall):** The conditional mean loss across the worst 5% tail scenarios.

5. Baseline Project Valuation & Results Scorecard

Results calculated for the baseline 100MW Solar PV Power Plant (\$30M CapEx, 20-Year Life):

Financial Performance Metric	Calculated Value	Target Hurdle / Benchmark	Financial Interpretation
Net Present Value (NPV)	+\$33.19 Million	> \$0.00 Million	Highly value-accretive; project viable
Internal Rate of Return (IRR)	18.52%	WACC = 8.00%	Generates +10.52% excess return spread
Levelized Cost of Energy (LCOE)	\$24.81 / MWh	Tariff = \$48.00 / MWh	Robust profit margin of +\$23.19 / MWh
Profitability Index (PI)	2.11x	> 1.00x	Every \$1.00 invested yields \$2.11 in PV
Discounted Payback Period	7 Years	< 20 Years Lifetime	Capital fully recovered in Year 7
Probability of Negative NPV	0.00% (Low Risk)	< 15.00% Threshold	Safe investment under 5,000 shocks
95% Value-at-Risk (VaR)	+\$18.42 Million	> \$0.00 Million	Even in worst 5% case, project is profitable

6. Mid-Review Viva Defense Cheat-Sheet

Q1: What is the core financial engineering problem solved by this project?

A: It evaluates whether multi-decade clean infrastructure investments are economically viable by discounting lifetime cash inflows against upfront capital costs using WACC, while stress-testing against market volatility via Monte Carlo simulation.

Q2: What is Levelized Cost of Energy (LCOE)?

A: LCOE is the net present value of all capital and operating costs divided by total discounted electricity output. It represents the exact floor price per MWh needed to break even.

Q3: Why compare IRR against WACC?

A: WACC is the cost of borrowing capital. If project IRR exceeds WACC, the asset generates returns higher than financing costs, creating net shareholder value.

Q4: Why implement Monte Carlo simulation if deterministic NPV is positive?

A: Deterministic NPV assumes static tariffs and zero cost overruns for 20 years. Monte Carlo simulates 5,000 joint stochastic combinations to quantify downside tail-risk and Value-at-Risk (VaR).

Q5: How does this fulfill all course guidelines?

A: It pulls and cleans real public infrastructure data, provides descriptive statistics, and delivers two distinct financial engineering analyses: Deterministic Capital Budgeting (NPV/IRR/LCOE) and Stochastic Monte Carlo Risk Modeling.